

One-Day-Ahead 1% VaR Forecasting on QQQ Daily Log Returns

A stepwise model-selection study

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1 Introduction

This report studies one-day-ahead 1% Value-at-Risk (VaR) forecasting for QQQ daily log returns. Rather than presenting a flat catalog of models, I organize the report as a modeling sequence: begin with simple baseline specifications, examine where they fall short, use that evidence to motivate the next improvement, and continue until a convincing final model emerges.

That structure is important because the central difficulty is not volatility forecasting alone. For 1% left-tail risk, the main issue is tail calibration. A model may track conditional variance reasonably well and still perform poorly for VaR if its lower-tail distribution is too thin. The empirical evidence shows exactly that pattern. Gaussian volatility-based baselines are natural starting points, but they underestimate downside risk for QQQ. Student- t innovations help, and asymmetric volatility helps somewhat, but the largest improvement comes from replacing fixed parametric tails with empirical tail calibration through filtered historical simulation (FHS) and closely related bootstrap methods.

The report therefore follows this path:

- start with **RiskMetrics** and **Gaussian GARCH(1,1)** as baseline volatility-VaR models,
- move to **Student- t GARCH(1,1)** once Gaussian tails prove too thin,
- consider two next directions: richer GARCH-family extensions and alternative nonparametric or direct-quantile approaches,
- then follow the strongest remaining path from Lecture 18 by using **empirical residual tails**,
- and arrive at **GJR-FHS** as the best final model, with **GARCH-FHS** a close second.

All models discussed below are evaluated on a common out-of-sample period of 848 trading days. When earlier notes and current empirical results differ, the discussion follows the current integrated results used in the report.

2 Historical Background

VaR became especially prominent in the 1990s as financial institutions looked for a compact measure of downside market risk. J.P. Morgan's *RiskMetrics* framework was a major practical milestone because it popularized daily variance-based VaR measurement using exponentially weighted volatility updating. That framework also shaped later classroom and industry treatments of market risk, which is why RiskMetrics remains a natural baseline in a study like this.

The econometric foundation behind modern VaR forecasting comes from conditional heteroskedasticity models. Engle's ARCH model and Bollerslev's GARCH extension formalized the idea that financial returns have time-varying conditional variance, with volatility clustering as a central empirical feature. Once volatility is modeled dynamically, one-step conditional quantiles can be constructed by combining the conditional mean, the conditional standard deviation, and an assumed or estimated innovation quantile.

As VaR became more widely used, evaluation became just as important as construction. Kupiec (1995) proposed the unconditional coverage test, which checks whether the observed violation rate matches the nominal level. Christoffersen (1998) extended this logic by showing that a satisfactory VaR model should not only produce the right number of violations on average, but should also avoid clustering those violations through time.

A second important development was the move away from purely Gaussian tails. Financial returns often exhibit heavier tails than Gaussian theory suggests, so researchers increasingly used Student- t innovations or semi-parametric approaches. Barone-Adesi, Giannopoulos, and Vosper (1999) proposed Filtered Historical Simulation, which combines a volatility model with the empirical distribution of standardized residuals. That idea is especially relevant here, because the main

empirical lesson of this report is that empirical tail calibration matters more than simply refining the volatility recursion.

A different line of work models the conditional quantile directly rather than modeling the full conditional distribution first. Engle and Manganelli's CAViaR framework is the leading example. It is theoretically attractive because VaR itself is the object of interest, but it also requires careful optimization and adequacy testing.

Finally, there is a broader literature on volatility models with exogenous regressors. In this report, that role is played by VIX in a GARCHX benchmark. Economically, that is plausible because VIX reflects market-implied uncertainty. Empirically, however, the question is whether it improves one-day QQQ VaR enough to justify the additional structure.

3 Purpose Statement and Mathematical Development

3.1 VaR and the forecasting objective

Let r_t denote the QQQ daily log return,

$$r_t = \log \left(\frac{P_t}{P_{t-1}} \right).$$

For tail probability $\alpha = 0.01$, the one-day-ahead VaR is the conditional lower-tail quantile satisfying

$$\Pr(r_{t+1} < \text{VaR}_{t+1}(0.01) \mid \mathcal{F}_t) = 0.01.$$

So a 1% VaR forecast is a threshold that should be breached on roughly 1% of future days if the model is well calibrated.

For many models in this report, returns are written in location-scale form,

$$r_{t+1} = \mu_{t+1} + \sigma_{t+1} z_{t+1},$$

which implies

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \hat{q}_\alpha.$$

This decomposition organizes the model comparison. Different methods differ in two ways:

1. how they forecast σ_{t+1} ,
2. how they obtain the tail quantile $\hat{q}_\alpha$.

The modeling sequence in Section 5 can therefore be read as a sequence of attempts to improve one or both parts of this forecast.

3.2 Rolling out-of-sample design

The empirical study uses:

- 2,824 cleaned observations,
- a 70% initial training split,
- 848 one-step-ahead out-of-sample forecasts,
- an expected number of 1% violations equal to 8.48.

At each forecast origin, the model is re-estimated using information available up to that date and then used to forecast next-day VaR. This rolling design is appropriate because VaR is a risk-management tool used sequentially in real time rather than an in-sample fitting exercise.

3.3 Violation indicator

Given a forecast $\widehat{\text{VaR}}_t$, define the violation indicator by

$$I_t = \mathbf{1}\{r_t < \widehat{\text{VaR}}_t\}.$$

If the model is correctly calibrated at the 1% level, the sequence I_t should have mean close to 0.01 and should not display systematic clustering.

4 Data Example

The study uses QQQ adjusted closing prices to construct daily log returns and uses VIX closing values only in the GARCHX benchmark. The aim of the data example is not to document the full workflow, but to show how the same return series can produce quite different one-day VaR forecasts depending on the modeling assumptions.

A representative forecast date is 2025-04-04, when QQQ experienced a large negative return. This date is useful because it highlights how the models respond in a stressed setting.

Date	Model	VaR	Return	Viol.
2025-04-04	GJR-FHS	-0.0796	-0.0641	No
2025-04-04	GARCH-FHS	-0.0706	-0.0641	No
2025-04-04	GARCH- t	-0.0631	-0.0641	Yes
2025-04-04	HS	-0.0366	-0.0641	Yes
2025-04-04	RM(0.94)	-0.0453	-0.0641	Yes
2025-04-04	GARCHX	-0.0491	-0.0641	Yes
2025-04-04	CAViaR	-0.0629	-0.0641	Yes

The table illustrates the broader comparison developed later in the report. The best empirical-tail models move much farther into the left tail during stress, while several baseline models remain too close to zero and are breached.

Representative implementation snippets are collected in **Appendix A** so that the main discussion can remain focused on the econometric ideas and empirical results.

5 Stepwise Model Development and Comparison

5.1 Baseline Gaussian volatility models

5.1.1 Why start here?

The natural starting point is the standard volatility-to-VaR pipeline used in many introductory treatments of market risk: first forecast tomorrow’s conditional variance, then map that forecast into VaR with a Gaussian quantile. The two baseline models are **RiskMetrics** and **Gaussian GARCH(1,1)**.

These are reasonable starting points for three reasons. First, they are historically important. Second, they are mathematically transparent. Third, they frame the first empirical question clearly: if volatility is modeled dynamically but the tail is still Gaussian, is that enough for 1% VaR on QQQ?

5.1.2 RiskMetrics ($\lambda = 0.94$)

RiskMetrics uses the EWMA recursion

$$\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2, \quad \lambda = 0.94,$$

with one-step VaR forecast

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \Phi^{-1}(\alpha).$$

Its role in this report is to serve as a simple, historically central benchmark.

5.1.3 Gaussian GARCH(1,1)

Gaussian GARCH(1,1) allows more flexible variance dynamics:

$$r_t = \mu + \varepsilon_t,$$

$$\varepsilon_t = \sigma_t z_t, \quad z_t \sim N(0, 1),$$

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2.$$

The one-step VaR mapping is again

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \Phi^{-1}(\alpha).$$

The intuition is straightforward: if volatility clustering is the main issue, then GARCH should outperform a simpler EWMA benchmark.

5.1.4 Results

Model	Viol.	Rate	Kupiec	Christ.	CC	Check loss
RM(0.94)	18	2.12%	0.0043	0.3925	0.0117	0.3781
GARCH-N	22	2.59%	0.0001	0.5970	0.0005	0.3700

Both baseline models produce too many breaches. The message from the first step is therefore clear: even when conditional volatility is updated dynamically, Gaussian lower tails are too thin for QQQ one-day 1% VaR. That finding motivates the next improvement: keep the GARCH volatility structure, but replace the Gaussian innovation law with a heavier-tailed parametric alternative.

5.2 5.2 Heavier-tailed parametric improvement: Student- t GARCH

5.2.1 Why is Student- t the next step?

Once the Gaussian baselines underperform, the most natural next step is not to abandon volatility modeling, but to allow the innovation distribution to have heavier tails. Student- t GARCH keeps the same GARCH variance recursion while replacing Gaussian innovations with a Student- t law.

5.2.2 Mathematical setup

The model is

$$r_t = \mu + \varepsilon_t,$$

$$\varepsilon_t = \sigma_t z_t, \quad z_t \sim t_\nu \text{ standardized to unit variance,}$$

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2.$$

The one-step VaR forecast becomes

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \hat{q}_\alpha^{(t)},$$

where $\hat{q}_\alpha^{(t)}$ is the fitted standardized Student- t 1% quantile.

5.2.3 Results

Model	Viol.	Rate	Kupiec	Christ.	CC	Check loss
GARCH- t	13	1.53%	0.1480	0.1890	0.1482	0.3503

Student- t GARCH clearly improves on the Gaussian baselines. The violation rate drops substantially, and check loss improves as well. That is direct evidence that tail thickness matters. Even so, the model still remains above the 1% target and does not yet provide the strongest overall calibration. Student- t innovations help, but they do not fully solve the forecasting problem.

At this stage, two natural directions remain:

1. enrich the **volatility model family**,
2. consider **simpler nonparametric or direct-quantile alternatives**.

5.3 5.3 Extension path 1: GARCH-family extensions

5.3.1 GARCHX(VIX)

5.3.1.1 Purpose and intuition

After Student- t GARCH, one natural question is whether the volatility forecast can be improved by adding market information external to QQQ returns. VIX is an obvious candidate because it summarizes market-implied uncertainty and may contain forward-looking information about near-term risk.

5.3.1.2 Mathematical setup

The benchmark can be written as

$$y_t = r_t - \mu,$$

$$\sigma_t^2 = \omega + \alpha y_{t-1}^2 + \beta \sigma_{t-1}^2 + \delta x_t,$$

where x_t is standardized VIX. The one-step VaR forecast is then

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \mu + \hat{\sigma}_{t+1|t} \Phi^{-1}(\alpha).$$

This is a package-based Gaussian-QML benchmark rather than a final empirical-tail model.

5.3.1.3 Results

Model	Viol.	Rate	Kupiec	Christ.	CC	Check loss
GARCHX	15	1.77%	0.0424	0.2621	0.0679	0.3549

The empirical evidence does not support pushing further in this direction. Although adding VIX is economically sensible, the model is weaker than plain Student- t GARCH and much weaker than the best empirical-tail models introduced later. GARCHX is therefore useful as a benchmark, but it is not the path that leads to the strongest final model.

5.3.2 GJR-GARCH

5.3.2.1 Purpose and intuition

Another natural extension is to enrich the variance recursion itself by allowing asymmetric volatility responses. In equity markets, negative returns often increase future volatility more than positive returns of the same size. GJR-GARCH is designed to capture that leverage effect.

5.3.2.2 Mathematical setup

The GJR-GARCH recursion is

$$r_t = \mu + \varepsilon_t,$$

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \gamma \varepsilon_{t-1}^2 I(\varepsilon_{t-1} < 0) + \beta \sigma_{t-1}^2.$$

If $\gamma > 0$, negative shocks raise future volatility more strongly than positive shocks.

The VaR mapping remains

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \hat{q}_\alpha,$$

with either Gaussian, Student- t , or later empirical residual quantiles.

5.3.2.3 Results before empirical-tail calibration

Model	Viol.	Rate	Kupiec	Christ.	CC	Check loss
GJR-GARCH	15	1.77%	0.0424	0.4621	0.0972	0.3450

GJR-GARCH does not yet solve the tail-calibration problem, but it is still a more promising extension than GARCHX. Its check loss is slightly better than plain Student- t GARCH, which suggests that asymmetry is useful even though it is not sufficient on its own.

5.4 5.4 Extension path 2: Historical / direct-quantile alternatives

5.4.1 Historical Simulation

5.4.1.1 Purpose and intuition

Historical Simulation is included as a deliberately simple nonparametric benchmark. Rather than estimating a volatility model or imposing a parametric innovation distribution, it asks whether the recent empirical distribution of returns can provide a competitive one-day tail forecast.

5.4.1.2 Mathematical setup

Using a rolling window of size $W = 250$,

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \text{Quantile}_{\alpha}(r_{t-W+1}, \dots, r_t).$$

5.4.1.3 Results

Model	Viol.	Rate	Kupiec	Christ.	CC	Check loss
HS	12	1.42%	0.2529	0.1569	0.1910	0.4092

Historical Simulation is a serious benchmark. It is not the best model in the comparison, but it is clearly competitive, which keeps the model-selection story honest: additional complexity must be justified empirically.

5.4.2 CAViaR-SAV

5.4.2.1 Purpose and intuition

CAViaR follows a different idea entirely. Instead of modeling volatility first and converting it into VaR, it models the conditional quantile directly.

5.4.2.2 Mathematical setup

The implemented Symmetric Absolute Value (SAV) recursion updates a positive latent risk level by

$$v_t = \beta_0 + \beta_1 v_{t-1} + \beta_2 |r_{t-1}|,$$

and maps back to the left-tail quantile through

$$q_t = -v_t.$$

Parameters are estimated by minimizing the quantile check-loss criterion,

$$L(\beta) = \sum_t \rho_\alpha(r_t - q_t(\beta)),$$

where

$$\rho_\alpha(u) = u(\alpha - I(u < 0)).$$

5.4.2.3 Scope of the implementation

The implementation used here is deliberately simplified. Only the SAV form is included, optimization is relatively light, and the Dynamic Quantile adequacy test from the original CAViaR framework is not part of the current study. It should therefore be interpreted as an exploratory direct-quantile benchmark rather than a full replication of the original paper.

5.4.2.4 Results

Model	Viol.	Rate	Kupiec	Christ.	CC	Check loss
CAViaR	6	0.71%	0.3664	0.0297	0.0625	0.3400

This is a nuanced result. The model is not a coverage failure: its unconditional coverage and check loss are numerically strong. However, the independence test fails, which means the timing of exceptions is not convincing enough. Combined with the simplified implementation, that makes the direct-quantile path interesting but not persuasive enough to become the main recommendation in this report.

5.5 5.5 Final improvement: empirical-tail FHS / bootstrap methods

By this point, the evidence suggests the following:

- Gaussian baselines are too thin-tailed.
- Student- t helps, but not enough.
- GARCHX is not a strong direction here.
- GJR-style asymmetry is more promising than GARCHX.
- Historical Simulation is a strong benchmark.
- Simplified CAViaR is interesting but not convincing enough as the main path.

That naturally leads to the final improvement from Lecture 18: keep the volatility model, but estimate the lower tail empirically rather than through a fixed parametric quantile.

5.5.1 Stage 1: apply empirical tails to plain GARCH

5.5.1.1 Purpose and intuition

Filtered Historical Simulation replaces the parametric innovation quantile with the empirical lower-tail quantile of standardized residuals. The idea is simple: let the volatility model handle scale dynamics, and let the residual data determine the tail.

5.5.1.2 Mathematical setup

After fitting a volatility model, compute standardized residuals

$$\hat{z}_t = \frac{r_t - \hat{\mu}_t}{\hat{\sigma}_t},$$

then set

$$\hat{q}_{\alpha, \text{emp}} = \text{empirical } \alpha\text{-quantile of } \{\hat{z}_t\}.$$

The one-step VaR is

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \hat{q}_{\alpha, \text{emp}}.$$

5.5.1.3 Results for plain GARCH empirical-tail variants

Model	Viol.	Rate	Kupiec	Christ.	CC	Check loss
GARCH-FHS	7	0.83%	0.5984	0.7327	0.8212	0.3499
GARCH- <i>t</i> -FHS	6	0.71%	0.3664	0.7698	0.6371	0.3611
GARCH-Boot	5	0.59%	0.1934	0.8075	0.4166	0.3523
GARCH- <i>t</i> -Boot	6	0.71%	0.3664	0.7698	0.6371	0.3648

This is the first major improvement in the report. Relative to Gaussian GARCH, empirical-tail GARCH variants move much closer to the desired 1% hit rate and pass the main backtests far more comfortably. The strongest message so far is that tail calibration contributes more than changing the volatility recursion alone.

5.5.2 Stage 2: apply empirical tails to GJR-GARCH

5.5.2.1 Purpose and intuition

If asymmetry added something useful earlier, the next question is whether asymmetry and empirical tails work even better together.

5.5.2.2 Mathematical setup

The variance recursion remains GJR-GARCH,

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \gamma \varepsilon_{t-1}^2 I(\varepsilon_{t-1} < 0) + \beta \sigma_{t-1}^2,$$

but the tail quantile is now empirical rather than Gaussian or Student- t :

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \hat{q}_{\alpha, \text{emp}}.$$

5.5.2.3 Results for GJR empirical-tail variants

Model	Viol.	Rate	Kupiec	Christ.	CC	Check loss
GJR-FHS	8	0.94%	0.8672	0.6961	0.9137	0.3402
GJR- t -FHS	6	0.71%	0.3664	0.7698	0.6371	0.3537

This final step completes the modeling sequence. Compared with plain GARCH-FHS, adding asymmetry through GJR produces the cleanest final calibration: the violation rate is closest to the nominal 1%, and the Kupiec, independence, and conditional coverage tests are all strong.

5.5.3 Main lesson from the stepwise path

The biggest overall gain comes from **empirical-tail calibration**, not just from making the volatility recursion more elaborate. Student- t helps. Asymmetry helps somewhat. But the decisive improvement arrives when the tail is estimated empirically. That is why the report ends with **GJR-FHS** as the best final model.

6 Backtesting and Forecast Evaluation Metrics

The model comparison in Section 5 uses four evaluation tools. Each should be defined before it is interpreted.

6.1 Kupiec unconditional coverage test

The null hypothesis is that the violation probability equals the target level α . If x is the number of violations in n forecasts and $\hat{\pi} = x/n$, the likelihood-ratio statistic is

$$LR_{uc} = -2 \log \left[\frac{(1 - \alpha)^{n-x} \alpha^x}{(1 - \hat{\pi})^{n-x} \hat{\pi}^x} \right].$$

A large p-value means the observed violation rate is plausible under the target 1% rate. A small p-value means the model is systematically too liberal or too conservative.

6.2 Christoffersen independence test

Correct hit frequency is not enough if violations arrive in clusters. The independence test treats the hit sequence as a first-order Markov chain and compares:

- a model where hits are independent through time,
- a model where hit probability depends on whether the previous day was a hit.

A low p-value indicates violation clustering and therefore poor dynamic calibration.

6.3 Conditional coverage test

The conditional coverage test combines the previous two ideas:

$$LR_{cc} = LR_{uc} + LR_{ind}.$$

It is the most natural summary test because a good VaR model should get both the overall hit rate and the timing pattern roughly right.

6.4 Check Loss

Because backtests based only on violations ignore the size of forecast errors, the report also uses quantile check loss. The formula used here is

$$\rho_\alpha(u) = u(\alpha - I(u < 0)), \quad u_t = r_t - \widehat{\text{VaR}}_t.$$

Equivalently,

$$\text{CheckLoss}_t = (r_t - \widehat{\text{VaR}}_t) (\alpha - I(r_t < \widehat{\text{VaR}}_t)).$$

For non-violation days, this is approximately $\alpha(r_t - \widehat{\text{VaR}}_t)$, so overly conservative forecasts still incur some loss. For violation days, it is approximately $(1 - \alpha)(\widehat{\text{VaR}}_t - r_t)$, so large missed tail events are penalized much more heavily. Lower check loss is therefore better.

This metric is useful because it evaluates the quality of the full quantile forecast rather than only whether the realized return barely crossed the threshold.

7 Forecast Uncertainty and 95% Confidence Intervals

7.1 Why point VaR forecasts need uncertainty measures

A VaR forecast is an estimate, not a known population quantity. It depends on estimated volatility, estimated or empirical tail quantiles, and finite-sample information. Reporting only a point forecast hides the fact that the forecast itself is uncertain.

7.2 General quantile-CI logic

A common large-sample approximation for a quantile standard error is

$$\text{SE}(\hat{q}_\alpha) = \sqrt{\frac{\alpha(1-\alpha)}{nf(\hat{q}_\alpha)^2}},$$

which leads to the generic 95% interval

$$\hat{q}_\alpha \pm 1.96 \text{ SE}(\hat{q}_\alpha).$$

For location-scale VaR models,

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \hat{q}_\alpha,$$

so quantile uncertainty maps directly into VaR uncertainty.

7.3 Model-specific CI constructions

The reported `lower_bound` and `upper_bound` values are not generated by one universal formula. They are model-specific.

7.3.1 Gaussian parametric models

For Gaussian models such as Gaussian GARCH, RiskMetrics, and the package-based GARCHX benchmark, the report uses the asymptotic standard error of the Gaussian quantile and maps it through

$$\hat{\mu} + \hat{\sigma} (\Phi^{-1}(\alpha) \pm 1.96 \text{ SE}(\hat{q}_\alpha)).$$

7.3.2 Student- t parametric models

For Student- t parametric models, the same logic is used but with the standardized Student- t density and quantile replacing the Gaussian ones.

7.3.3 FHS / empirical residual quantile models

For FHS models, the empirical α -quantile of standardized residuals is used, and quantile uncertainty is estimated from the residual sample before being mapped through the one-step volatility forecast.

7.3.4 Bootstrap models

For bootstrap models, standardized residuals are resampled, one-step returns are simulated, and the interval is built directly from the simulated predictive distribution.

7.3.5 Historical Simulation

Historical Simulation uses the same sample-quantile interval logic, but applied to the rolling window of recent raw returns rather than standardized residuals.

7.3.6 CAViaR-SAV proxy intervals

CAViaR does not produce a likelihood-based interval in the current specification. Instead, the procedure recenters an empirical return sample so that its 1% quantile matches the CAViaR point forecast and then applies sample-quantile interval logic to that shifted sample. This should be read as a proxy interval rather than the same inferential object used for parametric GARCH models.

7.4 Interval-width comparison

Table 1 reports the average widths of the reported 95% intervals.

Model	Mean CI width
CAViaR	0.02064
HS	0.02064
GARCH- t -FHS	0.00783
GARCH-FHS	0.00688
GARCH- t	0.00677
GJR-GARCH	0.00637
GJR- t -FHS	0.00586
GJR-FHS	0.00436
GARCH-N	0.00369
GARCH- t -Boot	0.00368
RM(0.94)	0.00368
GARCHX	0.00353
GARCH-Boot	0.00303

Three points are worth emphasizing. First, Historical Simulation and CAViaR have the widest intervals, which is consistent with the noisiness of empirical extreme quantiles and the proxy nature of the CAViaR interval construction. Second, the leading models combine moderate width with good calibration: GJR-FHS and GARCH-FHS do not simply minimize width, but instead balance stable interval behavior with strong backtesting performance. Third, narrow intervals are not automatically desirable. Several Gaussian-based models have relatively narrow intervals, but their VaR forecasts are too shallow, so narrowness alone is not evidence of superior risk forecasting.

8 Integrated Final Comparison Table

Table 2 summarizes the full model comparison. To keep the table readable, the columns focus on core statistics and the interpretation is given in the discussion immediately below.

Model	Viol.	Rate	Kupiec	Christ.	CC	Check loss	Label
GJR-FHS	8	0.94%	P	P	P	0.3402	Best
GARCH-FHS	7	0.83%	P	P	P	0.3499	Strong
GARCH- <i>t</i> -FHS	6	0.71%	P	P	P	0.3611	Strong
GJR- <i>t</i> -FHS	6	0.71%	P	P	P	0.3537	Strong
GARCH- <i>t</i> -Boot	6	0.71%	P	P	P	0.3648	Strong
GARCH-Boot	5	0.59%	P	P	P	0.3523	Cons.
HS	12	1.42%	P	B	P	0.4092	Bench.
GARCH- <i>t</i>	13	1.53%	P	P	P	0.3503	Mid
CAViaR	6	0.71%	P	F	B	0.3400	Mixed
GJR-GARCH	15	1.77%	F	P	B	0.3450	Mixed
GARCHX	15	1.77%	F	P	B	0.3549	Bench.
RM(0.94)	18	2.12%	F	P	F	0.3781	Weak
GARCH-N	22	2.59%	F	P	F	0.3700	Weak

Notes: P = pass ($p\text{-value} \geq 0.10$), B = borderline ($0.05 \leq p < 0.10$), F = fail ($p < 0.05$).

The table reinforces the main narrative. The strongest evidence favors **GJR-FHS**, which is closest to the nominal 1% target and performs well across the main backtests. **GARCH-FHS** is a close second and provides the simplest strong empirical-tail recommendation. The Student-*t* empirical-tail variants are also good models, but they are somewhat more conservative. Historical Simulation remains an important benchmark, while the simplified CAViaR result is best described as mixed: it is attractive on coverage and check loss but not on exception independence. At the bottom of the comparison are the Gaussian-tail baselines, especially Gaussian GARCH, which makes clear that volatility dynamics alone do not solve the left-tail forecasting problem.

9 Discussion

The empirical comparison points to a fairly clear methodological lesson. For one-day-ahead 1% VaR on QQQ, the hardest part is not simply forecasting volatility; it is calibrating the lower tail well once volatility has been forecast.

The first stage of the analysis shows that Gaussian volatility-to-VaR methods are too weak. RiskMetrics and Gaussian GARCH are natural baselines, but they systematically underestimate downside risk. The second stage shows that Student-*t* innovations improve matters in the expected direction, which is evidence that heavier tails are essential. The third stage shows that richer volatility structures alone are not the main answer. GARCHX is not persuasive in the current comparison, and GJR-GARCH helps only modestly. The fourth stage shows that Historical Simulation remains a serious benchmark and that direct-quantile modeling through simplified CAViaR is interesting but not yet convincing enough. The final stage shows that once the tail is estimated empirically, performance improves sharply.

The main substantive lesson is therefore simple: **the largest gain comes from empirical tail calibration, not just from changing the variance recursion.**

A second lesson is that asymmetry still matters, although less than tail treatment. GJR-FHS improves on GARCH-FHS by a modest but meaningful margin, which suggests that asymmetry is worth incorporating once the tail is handled properly.

10 Limitations

Several limitations should be stated directly.

1. **Integrated comparison from multiple model runs.** The final comparison is coherent because all included models share the same 848-day out-of-sample horizon, but the full set of results is assembled from more than one model run.
2. **GARCHX remains a benchmark.** The package-based implementation is cleaner than the earlier custom version, but its final VaR mapping is still Gaussian, so it should not be overinterpreted.
3. **The CAViaR implementation is simplified.** Only the SAV form is used, optimization is relatively light, and the Dynamic Quantile adequacy test from the original CAViaR framework is not included.
4. **Bootstrap forecasts are mechanically deterministic.** The current implementation resets the random seed inside the forecast function, which improves reproducibility but makes the rolling bootstrap less genuinely stochastic than a fresh-resampling implementation would be.
5. **The conclusions are application-specific.** The ranking is for QQQ, one-day horizon, and 1% VaR. Other assets, horizons, or tail levels could lead to different results.

11 Conclusion

This report reorganizes the VaR study as a stepwise model-selection exercise rather than a flat list of models. The starting point was a pair of Gaussian volatility-based VaR methods: **RiskMetrics** and **Gaussian GARCH(1,1)**. These were natural first choices because they are historically central, mathematically transparent, and closely aligned with standard course material. But the evidence showed that they were not enough. Both produced too many VaR breaches, indicating that Gaussian left tails are too thin for QQQ one-day 1% risk forecasting.

The next improvement was **Student- t GARCH(1,1)**. This helped, confirming that heavier-tailed innovations matter, but it did not fully solve the forecasting problem. After that, the report followed two directions. In the GARCH-family extension path, **GARCHX(VIX)** did not perform well enough to justify itself as the main direction, while **GJR-GARCH** suggested that asymmetry was somewhat useful. In the alternative-method path, **Historical Simulation** proved to be a strong benchmark, while the simplified **CAViaR-SAV** implementation remained interesting but not convincing enough because of dependence issues and implementation limits.

The final improvement came from the empirical-tail path emphasized in Lecture 18. Applying **FHS and related bootstrap ideas** to plain GARCH already improved performance substantially.

Applying the same idea to **GJR-GARCH** improved further and produced the best overall final model. The final recommendation is therefore **GJR-FHS**, with **GARCH-FHS** a close second.

The central methodological lesson is clear. For this application:

- Gaussian volatility-only VaR is too weak,
- Student- t helps,
- asymmetry helps somewhat,
- GARCHX is useful as a benchmark but not persuasive as the final path,
- Historical Simulation is a serious benchmark,
- the simplified CAViaR path remains interesting but not convincing enough,
- and the largest improvement comes from **empirical tail calibration**.

That is why the report ends with **GJR-FHS** as the best final model for one-day-ahead 1% VaR forecasting on QQQ daily log returns.

12 References

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13 Appendix A. Representative Implementation Snippets

This appendix collects short implementation snippets that illustrate how the main forecasts and evaluation measures are constructed. Only method-relevant code is included.

13.1 A.1 RiskMetrics recursion and VaR mapping

```
sigma2 = np.var(r_train[:250])
for i in range(250, len(r_train)):
    sigma2 = lam * sigma2 + (1 - lam) * r_train[i] ** 2

sigma_next = np.sqrt(sigma2)
var_t1 = r_train.mean() + sigma_next * norm.ppf(alpha)
```

13.2 A.2 GARCH one-step VaR with Student-*t* innovations

```
fit = arch_model(r_train * 100, p=1, q=1, dist="StudentsT").fit(dispatch="off")
mu_hat = fit.params.get("mu", 0.0) / 100
sigma_hat = np.sqrt(fit.forecast(horizon=1, reindex=False).variance.values[-1, 0]) / 100
nu_hat = fit.params["nu"]
q_alpha = t.ppf(alpha, df=nu_hat) * np.sqrt((nu_hat - 2) / nu_hat)
var_t1 = mu_hat + sigma_hat * q_alpha
```

13.3 A.3 GARCHX one-step forecast logic

```
y_adj = r_train - r_train.mean()
fit_garchx = garchx(y_adj, order=c(1, 1), xreg=vix_train)
pred_var = predict(fit_garchx, n.ahead=1, newxreg=vix_t)
sigma_next = sqrt(pred_var)
var_t1 = r_train.mean() + sigma_next * qnorm(alpha)
```

13.4 A.4 GJR-GARCH with empirical-tail mapping

```
fit = arch_model(r_train * 100, p=1, o=1, q=1, dist="normal").fit(dispatch="off")
mu_hat = fit.params.get("mu", 0.0) / 100
sigma_hat = np.sqrt(fit.forecast(horizon=1, reindex=False).variance.values[-1, 0]) / 100
z_hat = fit.std_resid.dropna().to_numpy()
q_emp = np.quantile(z_hat, alpha)
var_t1 = mu_hat + sigma_hat * q_emp
```

13.5 A.5 Historical Simulation

```
tail_sample = r_train[-250:]
var_t1 = np.quantile(tail_sample, alpha)
```

13.6 A.6 CAViaR-SAV recursion and check-loss objective

```
def sav_path(beta, r, v0):
    v = np.empty(len(r))
    v[0] = v0
    for t in range(1, len(r)):
        v[t] = beta[0] + beta[1] * v[t-1] + beta[2] * abs(r[t-1])
    return -v

u = r - q_path
loss = np.sum(u * (alpha - (u < 0).astype(float)))
```

13.7 A.7 FHS empirical residual quantile construction

```
fit = arch_model(r_train * 100, p=1, q=1, dist="normal").fit(dispatch="off")
mu_hat = fit.params.get("mu", 0.0) / 100
sigma_hat = np.sqrt(fit.forecast(horizon=1, reindex=False).variance.values[-1, 0]) / 100
z_hat = fit.std_resid.dropna().to_numpy()
q_emp = np.quantile(z_hat, alpha)
var_t1 = mu_hat + sigma_hat * q_emp
```

13.8 A.8 Bootstrap one-step simulation

```
boot_z = np.random.choice(z_hat, size=B, replace=True)
boot_returns = mu_hat + sigma_hat * boot_z
var_t1 = np.quantile(boot_returns, alpha)
```

13.9 A.9 Check-loss calculation

```
u = y - q
check_loss = u * (alpha - (u < 0).astype(float))
```

13.10 A.10 Backtesting logic

```
violation = (r_t < var_t).astype(int)
```